

Determining Solutions to Inequalities

MA.8.AR.2.2

I can decide whether a value is a solution, solve an inequality, and graph the solution set.

A new benchmark, and a grade 8 one.

Everything about equations still works. One rule is added.

Warm-Up

On your own. Four minutes. Boards, then paper.

Diagnostic question

Which of these is TRUE ?

1. $-4 < -6$
2. $-4 > -6$
3. $-4 = -6$
4. None — negatives cannot be compared.

Notes

I. A solution is any value that makes it true

- A. inequality — a statement that two expressions are not necessarily equal
- B. **A solution is any value that makes the statement TRUE when substituted.**
- C. **An inequality has infinitely many solutions, not one.**

Notes

II. The equal bar decides whether the boundary counts

- A. The symbols: $<$ less than, $>$ greater than, $\leq$ at most, $\geq$ at least.
- B. **With $\leq$ or $\geq$ the boundary value is itself a solution.**
- C. **With $<$ or $>$ it is not.**

Notes

III. Multiplying or dividing by a negative reverses it

- A. Add or subtract on both sides — nothing changes.
- B. Multiply or divide by a POSITIVE number — nothing changes.
- C. **Multiply or divide by a NEGATIVE — the symbol REVERSES.**
- D. **$4 < 6$ is true. Multiply by -1 and $-4 > -6$.**

Notes

IV. Open circle, closed circle, and which way to shade

- A. **OPEN** circle for $<$ and $>$. The boundary is not included.
- B. **CLOSED** circle for $\leq$ and $\geq$. The boundary is included.
- C. Shade the side your SOLVED line points to, not the side the original pointed to.
- D. If you are unsure, test one number on each side.

Example 1

Is it a solution? No solving.

$$4x - 5 < 7$$

Test $x = 2$, then $x = 3$.

Example 1

Both, substituted.

x = 2	x = 3
$4(2) - 5 = 3$	$4(3) - 5 = 7$
$3 < 7$ TRUE	$7 < 7$ FALSE
a solution	not a solution

Example 1

Change one symbol.

$4x - 5 < 7$	$4x - 5 \leq 7$
$x = 3$ gives $7 < 7$, FALSE	$x = 3$ gives $7 \leq 7$, TRUE
3 is NOT a solution	3 IS a solution

Example 2

Now solve it.

$$4x - 5 \leq 7$$

Does anything reverse? Answer before you start.

Example 2

No reversal.

$$4x - 5 \leq 7$$

$$4x \leq 12$$

added 5 — no reversal

$$x \leq 3$$

divided by POSITIVE 4 — no reversal

Example 2

Now change one sign.

$$-4x - 5 \leq 7$$

$$-4x \leq 12$$

added 5 — still no reversal

$$x \geq -3$$

divided by **NEGATIVE 4** — **REVERSED**

Example 3

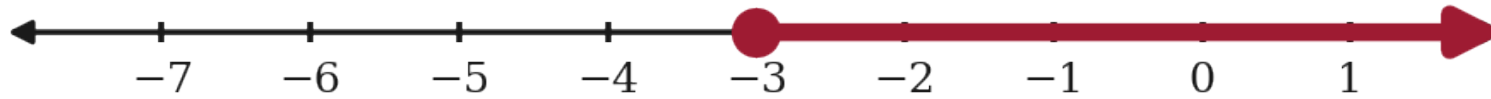
Graph $x \geq -3$.

Open circle or closed? And which way?

Say both before anybody draws anything.

Example 3

Closed circle, shaded right.



Example 3

Test it.

$x = 0$, inside the shading	$x = -5$, outside it
$-4(0) - 5 = -5$, and $-5 \leq 7$ ✓	$-4(-5) - 5 = 15$, and $15 \leq 7$ ✗

Example 4

Talia and Jen. Do not solve it yet.

Talia and Jen solved $8 - 1.2c \geq 5.6$. Talia says $c \leq -2$. Jen says $c \geq -2$.

Test value	Really a solution?	Talia says	Jen says
$c = -10$			
$c = 0$			
$c = 5$			

Example 4

Substituted.

Test value	Really a solution?	Talia says	Jen says
$c = -10$	yes	yes ✓	no ✗
$c = 0$	yes	no ✗	yes ✓
$c = 5$	no	no ✓	yes ✗

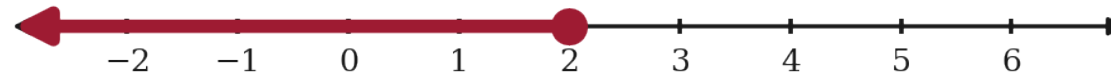
Example 4

Incomplete is not the same as wrong.

$$8 - 1.2c \geq 5.6$$

$$-1.2c \geq -2.4$$

$$c \leq 2$$



Whiteboards • Question 1 of 9

Boards up on three.

$4x - 5 < 7$ Is $x = 2$ a solution?

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 1 of 9

Answer.

$4x - 5 < 7$ Is $x = 2$ a solution?

$$4(2) - 5 = 3, \text{ and } 3 < 7$$

Answer: yes

Whiteboards • Question 2 of 9

Boards up on three.

$4x - 5 < 7$ Is $x = 3$ a solution?

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 2 of 9

Answer.

$4x - 5 < 7$ Is $x = 3$ a solution?

$4(3) - 5 = 7$, and $7 < 7$ is false

Answer: no

Whiteboards • Question 3 of 9

Boards up on three.

$4x - 5 \leq 7$ Is $x = 3$ a solution?

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 3 of 9

Answer.

$4x - 5 \leq 7$ Is $x = 3$ a solution?

$7 \leq 7$ is true

Answer: yes

Whiteboards • Question 4 of 9

Boards up on three.

Solve $4x - 5 \leq 7$

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 4 of 9

Answer.

Solve $4x - 5 \leq 7$

add 5, divide by POSITIVE 4

Answer: $x \leq 3$

Whiteboards • Question 5 of 9

Boards up on three.

Solve $-4x - 5 \leq 7$

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 5 of 9

Answer.

Solve $-4x - 5 \leq 7$

add 5, divide by NEGATIVE 4

Answer: $x \geq -3$

Whiteboards • Question 6 of 9

Boards up on three.

Graph $x \geq -3$

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 6 of 9

Answer.

Graph $x \geq -3$

closed circle, shaded right

Answer: see the board

Whiteboards • Question 7 of 9

Boards up on three.

-10.5

-5

-4.999

0

42

Which of these are solutions to $-25 < 4x - 5$?

More than one.

Whiteboards • Question 7 of 9

Answer.

−10.5 −5 −4.999 0 42

−25 < 4x − 5 means x > −5

Answer: −4.999, 0 and 42

Whiteboards • Question 8 of 9

Boards up on three.

Solve $5 + \frac{2}{3}b \geq 9$

Answer it.

Yes or no, a solution set, or a graph — read which.

Whiteboards • Question 8 of 9

Answer.

Solve $5 + \frac{2}{3}b \geq 9$

subtract 5: $\frac{2}{3}b \geq 4$, then multiply by $\frac{3}{2}$

Answer: $b \geq 6$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

$-3 - y/7 < -5$. Jonah says $y < -14$. Tanner says $y > -14$. Who is right?

Write your answer in sentences.

This one is written work. Justify it mathematically.

Whiteboards • Question 9 of 9

Answer.

$-3 - y/7 < -5$. Jonah says $y < -14$. Tanner says $y > -14$. Who is right?

add 3: $-y/7 < -2$, then $\times -7$ and REVERSE

Answer: Neither — $y > 14$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.